EC303 — Signals and System

Module 5: The Sampling Theorem & Signal Reconstruction

Prepared by **Biraj Sarkar (CGEC)** · MAKAUT B.Tech ECE (2023–27)

Authoritative Syllabus-Aligned Notes

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1. The Sampling Theorem

1.1 Significance and Concept

The sampling process forms the essential bridge connecting physical, continuous-time (analog) signals to discrete-time digital processors. The central question is: *Under what conditions can a continuous signal be represented by discrete samples without any loss of information, and how can it be reconstructed exactly?*

Nyquist-Shannon Sampling Theorem

A continuous-time bandlimited signal $x(t)$ with finite energy and maximum frequency component ω_m (or $f_m = \omega_m/2\pi$), such that:

$$X(j\omega) = 0 \quad \text{for all } |\omega| > \omega_m \tag{1}$$

can be uniquely, completely, and exactly reconstructed from its instantaneous samples $x(nT_s)$ taken at uniform time intervals T_s, if and only if the sampling frequency ω_s

satisfies:

$$\omega_s \geq 2\omega_m \iff f_s \geq 2f_m \tag{2}$$

where $f_s = 1/T_s$ is the sampling rate in Hertz, and $\omega_s = 2\pi/T_s$ is in radians per second.

1.2 Key Sampling Terminology

- **Nyquist Rate (f_N):** The absolute minimum sampling rate required to avoid information loss:

$$f_N \triangleq 2f_m \quad (\text{or } \omega_N \triangleq 2\omega_m) \tag{3}$$

- **Nyquist Interval (T_N):** The maximum permissible time interval between consecutive samples:

$$T_N \triangleq \frac{1}{f_N} = \frac{1}{2f_m} = \frac{\pi}{\omega_m} \tag{4}$$

- **Sampling Regimes:**
  1. *Over-sampling* ($f_s > 2f_m$)*:* Spectral replicas are separated by guard bands; exact reconstruction is easily achievable using practical filters.
  2. *Critical sampling* ($f_s = 2f_m$)*:* Spectral replicas touch at edges without overlap; requires an ideal, infinitely sharp brick-wall filter.
  3. *Under-sampling* ($f_s < 2f_m$)*:* Spectral replicas overlap, causing irreversible aliasing distortion.

2. Spectra of Sampled Signals (Impulse-Train Sampling)

2.1 Mathematical Model of Ideal Sampling

Ideal sampling is modeled as the product of the continuous-time analog signal $x(t)$ with a periodic Dirac impulse train $p(t)$:

$$p(t) \triangleq \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \tag{5}$$

The sampled signal $x_p(t)$ is given by:

$$x_p(t) = x(t) \cdot p(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s) \tag{6}$$

2.2 Spectral Derivation (Frequency Domain Analysis)

To determine the Fourier transform of $x_p(t)$, we first express the periodic impulse train $p(t)$ in its Exponential Fourier Series:

$$p(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_s t}, \quad \text{where } c_k = \frac{1}{T_s}\int_{-T_s/2}^{T_s/2} \delta(t)e^{-jk\omega_s t}dt = \frac{1}{T_s} \tag{7}$$

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Thus, the Fourier transform of the sampling train is:

$$P(j\omega) = \mathcal{F}\{p(t)\} = \frac{2\pi}{T_s}\sum_{k=-\infty}^{\infty}\delta(\omega - k\omega_s) = \omega_s\sum_{k=-\infty}^{\infty}\delta(\omega - k\omega_s) \tag{8}$$

Applying the multiplication property of the CTFT ($\mathcal{F}\{x_1(t)x_2(t)\} = \frac{1}{2\pi}[X_1(j\omega) * X_2(j\omega)]$):

$$X_p(j\omega) = \frac{1}{2\pi}[X(j\omega) * P(j\omega)] \tag{9}$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\theta)\left[\frac{2\pi}{T_s}\sum_{k=-\infty}^{\infty}\delta(\omega - \theta - k\omega_s)\right]d\theta \tag{10}$$

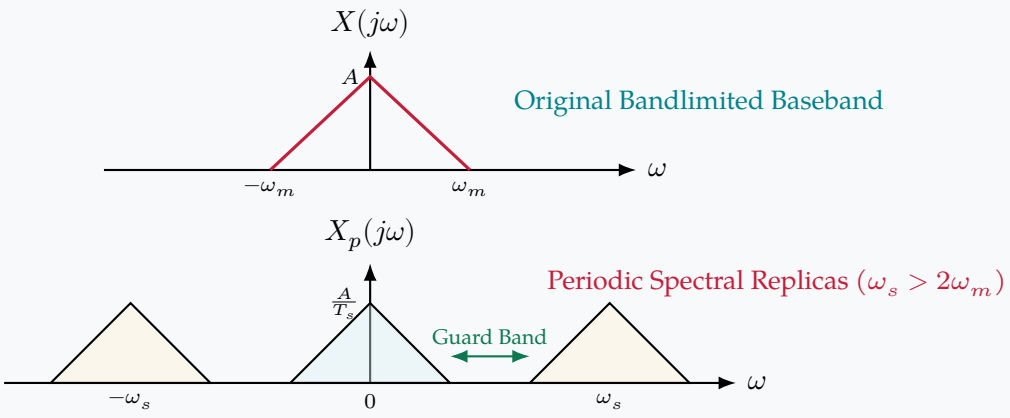
Interchanging integration and summation using the sifting property:

Core Formula: Spectrum of Sampled Signal

$$X_p(j\omega) = \frac{1}{T_s}\sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s)) \tag{11}$$

Physical Meaning: Sampling in the time domain causes the original continuous spectrum $X(j\omega)$ to be **periodically replicated across the frequency axis** at integer multiples of the sampling frequency ω_s, with each replica scaled by $1/T_s$.

TikZ Illustration: Spectra of Continuous, Impulse Train, and Sampled Signals



3. Aliasing and Anti-Aliasing Techniques

3.1 The Aliasing Phenomenon

When a signal is sampled below the Nyquist rate ($\omega_s < 2\omega_m$), the adjacent spectral replicas overlap:

$$\omega_s - \omega_m < \omega_m \implies \omega_s < 2\omega_m \tag{12}$$

In the overlap regions, spectral components from high-frequency replicas fold into the baseband:

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Definition: Aliasing

Aliasing is the spectral distortion phenomenon whereby high-frequency components of a continuous-time signal fold back across the Nyquist frequency ($f_s/2$) and appear masquerading as indistinguishable lower-frequency components in the sampled signal.

Demonstration: Aliased Frequency of a Sinusoid

Consider a continuous sinusoidal tone $x(t) = \cos(2\pi f_0 t)$ sampled at rate f_s:

$$x[n] = x(nT_s) = \cos(2\pi f_0 n/f_s)$$

If $f_0 = 70$ Hz and $f_s = 100$ Hz (where Nyquist frequency is $f_s/2 = 50$ Hz):

$$x[n] = \cos\left(2\pi\frac{70}{100}n\right) = \cos(1.4\pi n) = \cos((2\pi - 0.6\pi)n)$$
$$= \cos(-0.6\pi n) = \cos(0.6\pi n) = \cos\left(2\pi\frac{30}{100}n\right)$$

The 70 Hz signal appears identically as an apparent 30 Hz tone ($|f_0 - f_s| = |70 - 100| = 30$ Hz). The true high-frequency information is irrevocably corrupted.

3.2 Prevention of Aliasing: Anti-Aliasing Filter (AAF)

Once aliasing occurs, the overlapping spectral components sum together linearly and **cannot be separated** by any post-processing algorithm. To prevent aliasing:

1. **Pre-Filtering (Anti-Aliasing Filter):** An analog low-pass filter is inserted strictly *before* the sampler. The filter has a cutoff frequency $f_c \leq f_s/2$, suppressing any input spectral energy exceeding the Nyquist foldover limit.
2. **Over-sampling:** Selecting f_s significantly higher than $2f_m$ widens the transition band for the analog anti-aliasing filter, allowing lower-order, cost-effective analog filter designs.

4. Signal Reconstruction from Samples

4.1 The Ideal Reconstruction Filter (Whittaker-Shannon Interpolation)

When $\omega_s \geq 2\omega_m$, the original spectrum $X(j\omega)$ can be extracted from $X_p(j\omega)$ using an ideal low-pass filter with cutoff frequency ω_c ($\omega_m < \omega_c < \omega_s - \omega_m$) and gain T_s:

$$H_r(j\omega) = \begin{cases} T_s, & |\omega| \leq \frac{\omega_s}{2} \\ 0, & |\omega| > \frac{\omega_s}{2} \end{cases} \tag{13}$$

The impulse response $h_r(t)$ of this ideal reconstruction filter is:

$$h_r(t) = \mathcal{F}^{-1}\{H_r(j\omega)\} = \frac{T_s}{2\pi}\int_{-\omega_s/2}^{\omega_s/2} e^{j\omega t}d\omega = \frac{T_s}{2\pi}\left[\frac{e^{j\omega t}}{jt}\right]_{-\pi/T_s}^{\pi/T_s} = \frac{\sin(\pi t/T_s)}{\pi t/T_s} = \text{sinc}\left(\frac{t}{T_s}\right) \tag{14}$$

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Convolving the sampled impulse train $x_p(t)$ with $h_r(t)$ yields:

Whittaker-Shannon Cardinal Reconstruction Formula

$$x(t) = x_p(t) * h_r(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\text{sinc}\left(\frac{t - nT_s}{T_s}\right) \tag{15}$$

where $\text{sinc}(u) \triangleq \frac{\sin(\pi u)}{\pi u}$.

Geometric Interpolation Property: At the exact sampling instants $t = mT_s$:

$$\text{sinc}\left(\frac{mT_s - nT_s}{T_s}\right) = \text{sinc}(m-n) = \begin{cases} 1, & m = n \\ 0, & m \neq n \end{cases} \tag{16}$$

Thus, at each sampling point, the reconstructed signal equals the exact sample value $x(mT_s)$ because all neighboring sinc interpolation curves pass through zero.

Practical Limitation: The ideal sinc interpolator is **physically unrealizable** because $\text{sinc}(t/T_s)$ has infinite duration, requiring knowledge of all past and future samples ($h_r(t) \neq 0$ for $t < 0$, non-causal).

4.2 Zero-Order Hold (ZOH)

In practical digital-to-analog converters (DACs), the simplest and most ubiquitous hold device is the **Zero-Order Hold (ZOH)**, which clamps each sample value until the arrival of the next sample:

$$x_{zoh}(t) = x(nT_s) \quad \text{for } nT_s \leq t < (n+1)T_s \tag{17}$$

- **Impulse Response:** A causal rectangular pulse of duration T_s:

$$h_0(t) = u(t) - u(t - T_s) = \text{rect}\left(\frac{t - T_s/2}{T_s}\right) \tag{18}$$

- **Frequency Response:**

$$H_0(j\omega) = \int_0^{T_s} e^{-j\omega t}dt = \frac{1 - e^{-j\omega T_s}}{j\omega} = T_s\left(\frac{\sin(\omega T_s/2)}{\omega T_s/2}\right)e^{-j\omega T_s/2} = T_s\text{sinc}\left(\frac{\omega T_s}{2\pi}\right)e^{-j\omega T_s/2} \tag{19}$$

- **Distortion Effects of ZOH:**
  1. *In-band Droop (Sinc Roll-off):* Magnitude response attenuates at the band edge $\omega_s/2$ by a factor of $\text{sinc}(0.5) = \frac{2}{\pi} \approx 0.637$ (-3.92 dB).
  2. *High-Frequency Images:* The sinc side-lobes decay slowly ($\propto 1/\omega$), leaving residual harmonic images. A post-reconstruction smoothing filter and equalization circuit ($\frac{1}{\text{sinc}}$) are required.

4.3 First-Order Hold (FOH)

The **First-Order Hold (FOH)** performs linear extrapolation/interpolation between consecutive sample points.

- **Impulse Response (Triangular Pulse):**

$$h_1(t) = \begin{cases} 1 - \frac{|t|}{T_s}, & |t| \leq T_s \\ 0, & |t| > T_s \end{cases} \tag{20}$$

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- **Frequency Response:**

$$H_1(j\omega) = T_s \left(\frac{\sin(\omega T_s/2)}{\omega T_s/2} \right)^2 = T_s \text{sinc}^2 \left(\frac{\omega T_s}{2\pi} \right) \tag{21}$$

- **Advantage:** The side-lobes decay as $1/\omega^2$, providing vastly superior suppression of spectral images compared to ZOH. (A causal implementation requires 1 sample delay T_s).

5. Relation Between Continuous and Discrete-Time Systems

5.1 Mapping of Time and Frequency Variables

When a continuous signal $x_c(t)$ is sampled with period T_s, the discrete-time sequence is $x[n] = x_c(nT_s)$. The independent variables are related by:

$$t = nT_s = \frac{n}{f_s} \tag{22}$$

The relationship between continuous-time angular frequency ω (rad/s) and discrete-time angular frequency Ω (rad/sample) is:

$$\Omega = \omega T_s = \frac{\omega}{f_s} = 2\pi \frac{f}{f_s} \tag{23}$$

Key Frequency Correspondence:

- Continuous DC ($\omega = 0$) maps to Discrete DC ($\Omega = 0$).
- Continuous Nyquist frequency ($\omega = \omega_s/2 = \pi/T_s$) maps directly to the discrete half-sample frequency $\Omega = \pi$.
- Sampling frequency $\omega = \omega_s$ maps to $\Omega = 2\pi$ (which is identical to 0 due to 2π-periodicity).

5.2 Digital Processing of Continuous-Time Signals

A standard digital signal processing (DSP) system architecture comprises:

$$x_c(t) \xrightarrow{\text{AAF}} x_a(t) \xrightarrow{\text{C/D}} x[n] \xrightarrow{\boxed{H(e^{j\Omega})}} y[n] \xrightarrow{\text{D/C}} y_c(t)$$

If sampling satisfies the Nyquist condition ($f_s \geq 2f_m$), the overall continuous-time system behaves as an equivalent LTI analog filter with effective frequency response:

$$H_{eff}(j\omega) = \begin{cases} H(e^{j\Omega})\big|_{\Omega=\omega T_s}, & |\omega| < \frac{\pi}{T_s} \\ 0, & |\omega| \geq \frac{\pi}{T_s} \end{cases} \tag{24}$$

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Quick Revision & Examination Compendium

Module 5: The Sampling Theorem & Signal Reconstruction

Key Formulas at a Glance

Concept / Law	Definitive Mathematical Formula
Nyquist Sampling Criterion	$f_s \geq 2f_m \iff \omega_s \geq 2\omega_m$
Nyquist Rate	$f_N = 2f_m \quad (\text{Hz}), \quad \omega_N = 2\omega_m \quad (\text{rad/s})$
Nyquist Interval	$T_N = \frac{1}{2f_m} = \frac{\pi}{\omega_m} \quad (\text{seconds})$
Sampled Spectrum Replication	$X_p(j\omega) = \frac{1}{T_s}\sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s))$
Aliased Frequency	$f_{apparent} = \lvert f_0 - kf_s \rvert \leq \frac{f_s}{2}$
Ideal Sinc Interpolation	$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\text{sinc}\left(\frac{t-nT_s}{T_s}\right)$
Zero-Order Hold Transfer Function	$H_0(j\omega) = T_s\left(\frac{\sin(\omega T_s/2)}{\omega T_s/2}\right)e^{-j\omega T_s/2}$
First-Order Hold Transfer Function	$H_1(j\omega) = T_s\text{sinc}^2\left(\frac{\omega T_s}{2\pi}\right)$
CT to DT Frequency Mapping	$\Omega = \omega T_s = \frac{\omega}{f_s}$
Effective DSP Frequency Response	$H_{eff}(j\omega) = H(e^{j\omega T_s}) \quad \text{for } \lvert\omega\rvert < \frac{\pi}{T_s}$

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One-Line Definitions for Rapid Review

Term	Definitive Technical Meaning
Sampling Theorem	Principle stating a bandlimited signal can be perfectly recovered if sampled at $\geq 2\times$ maximum frequency.
Nyquist Rate	The minimum theoretical sampling frequency ($2f_m$) required to avoid spectral overlap and aliasing.
Aliasing	An irreversible distortion where high frequencies fold into lower bands due to inadequate sampling rates.
Anti-Aliasing Filter	An analog low-pass pre-filter that removes signal content above $f_s/2$ prior to sampling.
Whittaker-Shannon Interpolation	The mathematically exact reconstruction formula using weighted, time-shifted sinc functions.
Zero-Order Hold (ZOH)	A DAC reconstruction filter that holds each sample amplitude flat over the entire sampling interval.
First-Order Hold (FOH)	A reconstruction filter that performs linear triangular interpolation between consecutive sample points.
Guard Band	The frequency separation gap ($\omega_s - 2\omega_m$) between adjacent spectral replicas in over-sampled systems.
Anti-Imaging Filter	A post-DAC analog smoothing filter designed to suppress high-frequency periodic images from the hold circuit.

Frequently Asked / Viva Questions

Q1. What is the Nyquist rate for the composite signal $x(t) = 3\cos(100\pi t) + 2\sin(300\pi t) - 5\cos(200\pi t)$?

Identify the angular frequencies: $\omega_1 = 100\pi \implies f_1 = 50$ Hz, $\omega_2 = 300\pi \implies f_2 = 150$ Hz, $\omega_3 = 200\pi \implies f_3 = 100$ Hz. The highest frequency is $f_m = 150$ Hz. Thus, Nyquist rate $f_N = 2f_m = 2 \times 150 = 300$ Hz (or $\omega_N = 600\pi$ rad/s).

Q2. What is the Nyquist rate for $x(t) = \mathbf{sinc}(100t) \cdot \mathbf{sinc}(200t)$?

In the frequency domain, multiplication in time corresponds to convolution: $X(j\omega) = \frac{1}{2\pi}[X_1(j\omega) * X_2(j\omega)]$. The bandwidths of the rectangular spectra add: $W = W_1 + W_2 = 100\pi + 200\pi = 300\pi$ rad/s. Thus, $f_m = 150$ Hz, giving Nyquist rate $f_N = 2f_m = 300$ Hz.

Q3. Why cannot an aliased signal be corrected or de-aliased digitally after sampling?

Once aliasing occurs, overlapping high- and low-frequency spectral components sum together into identical frequency bins in the sampled sequence. Since a single composite sum contains no tag indicating individual source components, separation is mathematically impossible.

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Frequently Asked / Viva Questions (contd.)

Q4. Why is an Anti-Aliasing Filter placed before the sampler rather than after?

Because aliasing occurs during the sampling operation itself. Placing the filter after the sampler attempts to filter a signal that is already contaminated with folded noise that now resides in the legitimate signal band.

Q5. Why is ideal sinc interpolation practically unrealizable?

Because $\text{sinc}(t/T_s) = \frac{\sin(\pi t/T_s)}{\pi t/T_s}$ has infinite duration extending from $t = -\infty$ to $+\infty$. Realizing it would require infinite delay and infinite non-causal memory.

Q6. What is the passband droop caused by a Zero-Order Hold (ZOH)?

The ZOH frequency response has magnitude $|H_0(j\omega)| = T_s \left|\frac{\sin(\omega T_s/2)}{\omega T_s/2}\right|$. At the band edge $\omega = \omega_s/2$, $|H_0| = T_s \frac{\sin(\pi/2)}{\pi/2} = \frac{2}{\pi} T_s \approx 0.637 T_s$. This produces an in-band attenuation of $20\log_{10}(2/\pi) \approx -3.92$ dB, requiring an inverse-sinc equalizer.

Q7. Compare ZOH and FOH in terms of spectral image suppression.

ZOH has side-lobes that decay at a rate of $1/\omega$ (-20 dB/decade), whereas FOH has side-lobes that decay at $1/\omega^2$ (-40 dB/decade). FOH provides vastly superior attenuation of high-frequency images.

Q8. What continuous frequency ω corresponds to $\Omega = \pi$ in a discrete-time system?

$\Omega = \omega T_s = \pi \implies \omega = \frac{\pi}{T_s} = \frac{\omega_s}{2} = \pi f_s$. This corresponds precisely to the continuous-time Nyquist foldover frequency.

Q9. What is meant by flat-top sampling and how does it differ from natural sampling?

In natural sampling, the tops of the sampled pulses follow the exact analog contour of $x(t)$. In flat-top sampling, each pulse has a constant (flat) amplitude equal to the sample value at the pulse onset. Flat-top sampling introduces aperture distortion (sinc attenuation) similar to a sample-and-hold circuit.

Q10. What is the Nyquist rate for $x(t) = \cos^2(400\pi t)$?

Using the trigonometric identity: $\cos^2(\theta) = \frac{1+\cos(2\theta)}{2}$. Thus, $x(t) = \frac{1}{2} + \frac{1}{2}\cos(800\pi t)$. The maximum frequency is $f_m = 400$ Hz. Hence, the Nyquist rate is $f_N = 2f_m = 800$ Hz.

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Summary Matrix: Comparison of Signal Reconstruction Methods

Reconstruction Filter	Impulse Response $h(t)$	Frequency Response	Practical Feasibility
Ideal Sinc Interpolator	$\textbf{sinc}(t/T_s)$	**Ideal brick-wall boxcar**	Unrealizable; non-causal, infinite delay.
Zero-Order Hold (ZOH)	$\textbf{rect}\left(\frac{t-T_s/2}{T_s}\right)$ **(Staircase)**	$T_s\textbf{sinc}\left(\frac{\omega T_s}{2\pi}\right)e^{-j\omega T_s/2}$	Universal in DACs; causes -3.92 dB droop.
First-Order Hold (FOH)	$\textbf{tri}\left(\frac{t}{T_s}\right)$ **(Linear interpolation)**	$T_s\textbf{sinc}^2\left(\frac{\omega T_s}{2\pi}\right)$	Causal with 1 sample delay; -40 dB/dec rolloff.

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